

ERDŐS PROBLEM 617: AN INDEPENDENT CHECK OF THE FIXED CASES $r = 6$ AND $r = 7$

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SCOPE AND SOURCES

We audited, line by line, R. Sneiderman’s manuscripts for $r = 6$ (`r6/main.tex`, dated 18 July 2026) and for $r = 7$ (`r7-r8/main.tex`, Sections 2 and 3–4), from the repository `Robby955/erdos-617-fixed-cases`, commit `fb628c8` (25 July 2026). The case $r = 8$, the case $r = 9$ and the SAT certificates were *not* audited. The audit was done by an AI system (Claude) working with a human partner. It is *not* an expert referee report. Every computation used was small (graphs on at most 7 vertices, and integer recursions for $r = 7$). Verdict. Beyond the written repair already posted by `nwinter`’s agents (see Prior work), we found **no mathematical error in the parts we checked**. Every inequality and case split of the human-readable chain was redone by hand; the recursion values and the finite lemma were recomputed by machine only. We found one minor expository gap (Section 1). We also make three observations: Observations 1 and 2 show that neither proof needs the Kang–Pikhurko equality characterization, and Lemma 2 gives an independent short proof of the level-3 exclusion, which removes the need for Theorem 2.2 and for $\omega \leq r - 1$ at level 3.

PRIOR WORK (CHECKED 6 OCTOBER 2026)

- $r = 6$ was already reviewed by Nick Winter’s project (`nwinter/erdos-617-r5`, `reviews/sneiderman-r6.md`, 29–30 July 2026; verdict SURVIVES), and the $r = 6$ theorem was formalized there in Lean 4. That review already contains *both* of our $r = 6$ points: the redundancy of the endpoint lemma (addendum of 30 July, also found by the project’s second AI team, GPT-5.6-Sol-based and called “team-sol” in that repository) and the missing “nonbipartite” line (its §8.3). Our Observation 1 and the expository gap below are therefore **independent confirmations, not new**.
- $r = 7$: the proof-claims thread (read in full at erdosproblems.com/617 on 6 Oct. 2026) shows no comment on the $r = 7, 8$ claim. A comment by `nwinter`’s agents on the $r = 9$ claim (31 July) screens $r = 7, 8, 9$: “PASS after required written repair”. It flags (i) the inference “all

old parts have order r ” at `r7-r8/main.tex` line 403, with a three-step repair, and (ii) the eight-set-cap sentence at line 642, which belongs to the $\delta = 7$ branch only. It also states that the $r = 7$ enumeration was verified in full. **Our pass did not flag (i) or (ii)**; on re-reading we confirm both. Our Observation 2 (the passage is not load-bearing at $r = 7$) was not found in the sources consulted. Their full screen document is not public, so we do not call it new.

1. THE CASE $r = 6$

Checked by hand.

- Lemmas 2.4 (neighbourhood accounting), 2.5 (averaging), 2.6 (clique peeling), 2.7 (terminal complement).
- Proposition 3.1 (P_3): the table $d = 0, \dots, 4$ (lower bounds 94, 81, 71, 62, 56); the case $d = 5$ (the bound $1 \leq m \leq a \leq 4$, the cover argument with $\tau(L) \geq 7$, the three sub-cases $m \leq 2$, $m = 3$, $m = 4$); the case $d = 6$ (equality forcing $Y = X = 10$, $e(L) = 26$, and the classification of 5-edge graphs on 6 vertices, which forces $M = K_{1,5}$).
- Proposition 4.1 (19 vertices): the bounds 108, 95, 83, 74; $d = 4$ (forced isolated K_5 , a 5-regular L , then Lemma 2.7); $d = 5$ ($m + z \leq 5$, cover of size $\leq 6 < 8$); $d = 6$. In the case $d = 6$ we checked every degree-sequence sub-case: triangle-free M with $m = 5, \dots, 9$, and the case where M has a triangle, which ends at the forced weights $(5, 5, 2)$ and a K_6 .
- Propositions 5.1, 5.2 and 6.1: every averaging fraction in the three tables was recomputed exactly; the peeling chain $31 \rightarrow 25 \rightarrow 19 \rightarrow 13$ with edge bounds 96, 81, 66, 51; the final step at 13 vertices.
- Main proof: $e(G) \leq 111$, $\delta(G) \leq 5$ by Brooks, and the last table (all entries < 96 ; the case $d = 5$ gives exactly 96).

Observation 1 (Lemma 2.3 is not needed). Lemma 2.3 (the 18-vertex end-point) is the only place where the $r = 6$ proof uses the equality characterization of Kang–Pikhurko. Its conclusion $e(R) \geq 51$ is used only in the case $d = 5$ of Proposition 3.1. There it already follows from the accounting:

$$e(R) = d + (X + Y) + e(R[U]) \geq 5 + \binom{5}{2} + 36 = 51.$$

Here $X + Y \geq \binom{5}{2}$ comes from Lemma 2.4, and $e(R[U]) \geq 66 - 30 = 36$ comes from $\Delta(L) \leq 5$ on 12 vertices. So Lemma 2.3 can be deleted, together with the “three-partite” branch at the start of the proof of Proposition 3.1. The whole $r = 6$ proof then rests on three external facts:

- Brooks’s theorem;
- the bound $e(L) \leq \lfloor (n - 1)^2 / 4 \rfloor + 1$ for non-bipartite triangle-free graphs (the case $q = 2$ of Kang–Pikhurko, used once, with $n = 11$);
- elementary counting.

We note also that this proof uses no multi-colour density bound: only the one-colour consequences of the hypothesis (local cap, $\alpha \leq 6$, $e(G) \leq 111$). Minor expository gap ($r = 6$). In Proposition 3.1, case $d = 6$, the sentence “ L is triangle-free and nonbipartite” is not justified in the text. It follows at once from $\alpha(L) \leq 5$: a bipartition of 11 vertices has a side of order ≥ 6 .

2. THE CASE $r = 7$

Checked by hand.

- The shared setup: the local cap; the degree window (Brooks); the strict coloured density (7) via Kang–Pikhurko applied to each other colour.
- Theorem 2.1 (terminal core). At $n = 2r = 14$ it uses Andrásfai–Erdős–Sós: L is 6-regular and $6 > 2 \cdot 14/5$.
- Theorem 2.2 (exact ladder). We re-derived the linear form: the t -coefficient is A_s and the constant term is $-\widehat{B}_s$. We recomputed the thresholds by hand:

$$A_3 = 23, \widehat{B}_3 = -3 \Rightarrow T_3 = 1; \quad A_4 = 27, \widehat{B}_4 = 48 \Rightarrow T_4 = 2; \quad A_5 = 31, \widehat{B}_5 = 127 \Rightarrow T_5 = 5.$$

- Proposition 2.3 (block exclusion), including representative selection.
- Proposition 2.4 (soundness of the recursion), including the averaging bound $Y \leq \frac{r-2}{r} \binom{\delta}{2}$ behind $C_r(\delta)$.
- The star-and-cover mechanism; Lemma 3.3 ($P_4(28) \geq 100$, $P_4(29) \geq 113$); Lemma 3.4 ($P_5(35) \geq 122$, $P_5(36) \geq 134$, $P_6(43) \geq 156$, every case of δ); and the main chain $154 \rightarrow 133 \rightarrow 112$, which forces three disjoint K_7 , against block exclusion.
- The input floors $P_3(20) \geq 63$ and $P_3(21) \geq 75$, through the recursion. For example, $P_3(21) = \max(41 + 34, \lceil 147/2 \rceil)$ at $\delta = 7$, with $B_7(2, 13) = 41$ from Kang–Pikhurko and $C_7(7) = 34$.

Reproduced independently by computation (small).

- The margin table $B_r - E_{r,d,j}$ for $r = 7$ (the finite entries are 57, 42, 25, 18, 28, 10, 9, 5, 4 for $d \leq 5$, and $-1, -2, -3$ for $d = 6$; every $+\infty$ entry comes from the hand-checked thresholds), and the input floors $P_5(37) \geq 143$, $P_5(38) \geq 153$, $P_5(39) \geq 176$. We used our own reimplementations of the recursion, written from the consolidated draft and not from the author’s code. All values agree exactly with the author’s.
- The finite “weighted light edge” lemma (Lemma 3.2). We used an independent reimplementations of the idea behind the author’s second (Sage/nauty) route, with different tools: the networkx atlas of all unlabelled graphs on 7 vertices, with labelled counts $7!/|\text{Aut}|$. It reproduces the type counts 40, 65, 97, 131, the labelled counts 54, 257, 116, 280, 203, 490, 293, 930, and the full table of extrema; parts (i) and (ii) hold.

Observation 2 (no equality case needed for $r = 7$). The third line of Q_r (the “+1” gained by excluding the Kang–Pikhurko equality graphs when $m \geq ar + 1$) is the only use of the equality characterization. We replaced $\lfloor m/a \rfloor$ by $\lfloor m/a \rfloor - 1$ in that line and recomputed. All values used in the $r = 7$ proof are unchanged: the margin table and the floors 63, 75, 143, 153, 176. In fact every $B_7(a, m)$ with $a \leq 6$, $m \leq 49$ is unchanged. So the $r = 7$ proof needs only the Kang–Pikhurko *inequality*. (In passing, relying on Sneiderman’s description of the equality construction: under $\omega \leq r - 1$, an old part of order $\geq r$ in an equality graph already gives a forbidden clique. So the equality exclusion holds as soon as $m - 1 > a(r - 1)$, which is simpler than the argument given.)

3. A SELF-CONTAINED PROOF OF THE LEVEL-3 EXCLUSION

The $r = 7$ proof uses $T_3(7) = 1$ in Proposition 2.3, in Lemma 3.3, inside Theorem 2.2 (through $\Delta_4 = 3r + T_3 - 1$, which feeds T_4 and T_5) and in the recursion ($B_7(3, m) = +\infty$ for $m \geq 22$). Lemma 2 below proves a stronger statement (no hypothesis $\omega \leq r - 1$), so it can stand in at every one of these places. Lemma 2 uses the same method as Sneiderman’s Theorem 2.2. Lemma 1 uses a slightly different route from his Theorem 2.1: the non-bipartite triangle-free bound and the local cap replace Andrásfai–Erdős–Sós and $\omega \leq r - 1$. But Lemma 1 needs $|W| \geq 2r + 1$, so it does *not* replace Theorem 2.1 at $|W| = 2r = 14$, which Lemma 3.3 uses. Neither lemma is a new method; their value is an independent, short certificate. Throughout, G_i is any colour graph of a hypothetical counterexample.

Lemma 1 (Level 2). *Let $r \geq 4$. If $W \subseteq V$ and $\alpha(G_i[W]) \leq 2$, then $|W| \leq 2r$.*

Proof. Suppose $|W| = 2r + 1$ (restrict if larger) and put $L = \overline{G_i[W]}$. Then L is triangle-free. It is not bipartite: a bipartition would split W into two cliques of G_i , one of order $\geq r + 1$, and that clique violates the local cap. By the $q = 2$ case of Kang–Pikhurko, $e(L) \leq r^2 + 1$. By the coloured density bound, $e(L) \geq (r - 1)p_r(2r + 1) = (r - 1)(r + 2) = r^2 + r - 2$. This exceeds $r^2 + 1$ for $r > 3$. $\square$

Lemma 2 (Level 3). *Let $r \geq 6$. If $W \subseteq V$ and $\alpha(G_i[W]) \leq 3$, then $|W| \leq 3r$.*

Proof. Suppose $|W| = 3r + 1$ and put $L = \overline{G_i[W]}$, which is K_4 -free.

- *Upper bound.* For every v , the graph $L[N_L(v)]$ is triangle-free. Hence $\alpha(G_i[N_L(v)]) \leq 2$, and Lemma 1 gives $|N_L(v)| \leq 2r$. So $e(L) \leq (3r + 1)r$.
- *Lower bound.* Since $|W| > 3r$ and $\alpha \leq 3$, the graph $G_i[W]$ is not r -colourable. For each $j \neq i$, the graph $\overline{G_j[W]} \supseteq G_i[W]$ is therefore K_{r+1} -free and not r -partite. Kang–Pikhurko with $q = r$ and $n = 3r + 1$ gives $e(G_j[W]) \geq p_r(3r + 1) + 3 - 1 = 3r + 5$. Summing over the $r - 1$ colours $j \neq i$ gives $e(L) \geq (r - 1)(3r + 5) = 3r^2 + 2r - 5$.

- *Conclusion.* $3r^2 + 2r - 5 > 3r^2 + r$ for $r \geq 6$, a contradiction.

□

Remark. Lemma 2 replaces $T_3(r) = 1$ wherever it is used for $r \geq 6$; in particular, in the $r = 7$ proof. It is not a substitute for $T_4(7) = 2$ and $T_5(7) = 5$. These two values come from the ladder argument, whose derivation we re-checked by hand (above). The same “maximum degree + (7)” argument at level s , without $\omega \leq r - 1$, leaves the gap $\Phi_r(s) = -r(s^2 - 4s + 1) - 4s + 2$ on $2e(L)$. This is positive for $s = 3$ ($2r - 10$) and negative for $s \geq 4$ ($-r - 14$, $-6r - 18$, ...). That is why levels $s \geq 4$ need the induction through the thresholds T_{s-1} .

4. WHAT REMAINS TRUSTED

- For $r = 6$, the external facts are Brooks’s theorem and the non-bipartite triangle-free bound (by Observation 1). Nothing computational.
- For $r = 7$, the published external theorems are Kang–Pikhurko (inequality only, by Observation 2), Andrásfai–Erdős–Sós, Brooks and Turán.
- The computational inputs of $r = 7$ are:
 - the finite 7-vertex lemma (machine-checked only), checked by the author’s two programs and by our independent one;
 - the recursion values (the 9 finite margin entries for $d \leq 5$, plus $P_5(37)$, $P_5(38)$, $P_5(39)$), computed by the author’s code and reproduced by our independent reimplementation, but *not* checked by hand.
- This audit is not a substitute for expert review. In particular, we did not re-read the Kang–Pikhurko paper itself.